

Supplement to Six Birds Foundations VI: A Catalog of Dynamical Structural Laws

Ioannis Tsiokos 

6 October 2026 · Preprint v2.0

S1 What this supplement holds

This supplement accompanies *Six Birds Foundations VI: A Catalog of Dynamical Structural Laws*. The main paper states and proves every result; its numbered items are cited here by their numbers there. The supplement gives the Lean counterpart of each numbered item (Section S2), the case analysis and instance list of each law (Section S3), the register of non-implications with their separating examples (Section S5), the finite computations (Section S6), the axioms used by the Lean development (Section S7), the problems the laws come from (Section S8) and the commands that rebuild everything (Section S9).

Every declaration named in Sections S2 and S7 is present at the file and line given.

S2 Source concordance

Table S1 lists every numbered result of the main paper with its law, its part of the law, the line of the theorem catalog where that law’s statement begins, and its Lean declarations with their kind, file and line. Declaration names omit the common prefix of the library. Items argued on paper only are marked as such. Table S2 gives, for each law, the status recorded in the catalog and the status under which the main paper states it; where they differ, the main paper’s reading is the one supported by the Lean statement.

Table S1. Source concordance.

Item	Law and part	Lean declarations
Definition 6.1	G1, instance catalog line 155	<code>Instances.Collatz.exponent</code> (def) <code>lean/SixBirdsFoundationsVI/Instances/Collatz.lean:33</code> <code>Instances.Collatz.accelerated</code> (def) <code>lean/SixBirdsFoundationsVI/Instances/Collatz.lean:34</code> <code>Instances.Collatz.orbit</code> (def) <code>lean/SixBirdsFoundationsVI/Instances/Collatz.lean:61</code> <code>Instances.Collatz.A</code> (def) <code>lean/SixBirdsFoundationsVI/Instances/Collatz.lean:68</code> <code>Instances.Collatz.B</code> (def) <code>lean/SixBirdsFoundationsVI/Instances/Collatz.lean:73</code>
Theorem 6.2	G1, instance catalog line 155	<code>Instances.Collatz.accelerated_step</code> (theorem) <code>lean/SixBirdsFoundationsVI/Instances/Collatz.lean:36</code> <code>Instances.Collatz.affine_ledger</code> (theorem) <code>lean/SixBirdsFoundationsVI/Instances/Collatz.lean:78</code> <code>Instances.Collatz.accelerated_affine_ledger</code> (theorem) <code>lean/SixBirdsFoundationsVI/Instances/Collatz.lean:99</code>

Item	Law and part	Lean declarations
Theorem 6.3	G1, instance catalog line 155	Instances.Collatz.c_one_or_two (theorem) lean/SixBirdsFoundationsVI/Instances/Collatz.lean:119 Instances.Collatz.first_formula (theorem) lean/SixBirdsFoundationsVI/Instances/Collatz.lean:129 Instances.Collatz.split_pair (theorem) lean/SixBirdsFoundationsVI/Instances/Collatz.lean:151 Instances.Collatz.split_pair_next_exponent (theorem) lean/SixBirdsFoundationsVI/Instances/Collatz.lean:177 Instances.Collatz.split_pair_prior_exponent (theorem) lean/SixBirdsFoundationsVI/Instances/Collatz.lean:183
Corollary 6.4	G1, instance catalog line 155	Instances.Collatz.split_pair_next_exponent (theorem) lean/SixBirdsFoundationsVI/Instances/Collatz.lean:177 Instances.Collatz.split_pair_prior_exponent (theorem) lean/SixBirdsFoundationsVI/Instances/Collatz.lean:183
Remark 6.5	G1, instance catalog line 155	argued on paper; not formalized
Theorem 4.6	G5, instance catalog line 749	Instances.BinaryReverseAdd.transition_01 (theorem) lean/SixBirdsFoundationsVI/Instances/BinaryReverseAdd.lean:214 Instances.BinaryReverseAdd.transition_12 (theorem) lean/SixBirdsFoundationsVI/Instances/BinaryReverseAdd.lean:222 Instances.BinaryReverseAdd.transition_23 (theorem) lean/SixBirdsFoundationsVI/Instances/BinaryReverseAdd.lean:230 Instances.BinaryReverseAdd.transition_30 (theorem) lean/SixBirdsFoundationsVI/Instances/BinaryReverseAdd.lean:238 Instances.BinaryReverseAdd.word_transition_01 (theorem) lean/SixBirdsFoundationsVI/Instances/BinaryReverseAdd.lean:435 Instances.BinaryReverseAdd.word_transition_12 (theorem) lean/SixBirdsFoundationsVI/Instances/BinaryReverseAdd.lean:439 Instances.BinaryReverseAdd.word_transition_23 (theorem) lean/SixBirdsFoundationsVI/Instances/BinaryReverseAdd.lean:443 Instances.BinaryReverseAdd.word_transition_30 (theorem) lean/SixBirdsFoundationsVI/Instances/BinaryReverseAdd.lean:447 Instances.BinaryReverseAdd.p0_not_palindrome (theorem) lean/SixBirdsFoundationsVI/Instances/BinaryReverseAdd.lean:56 Instances.BinaryReverseAdd.p1_not_palindrome (theorem) lean/SixBirdsFoundationsVI/Instances/BinaryReverseAdd.lean:253 Instances.BinaryReverseAdd.p2_not_palindrome (theorem) lean/SixBirdsFoundationsVI/Instances/BinaryReverseAdd.lean:65 Instances.BinaryReverseAdd.p3_not_palindrome (theorem) lean/SixBirdsFoundationsVI/Instances/BinaryReverseAdd.lean:258 Instances.BinaryReverseAdd.family_closed (theorem) lean/SixBirdsFoundationsVI/Instances/BinaryReverseAdd.lean:472 Instances.BinaryReverseAdd.family_target_free (theorem) lean/SixBirdsFoundationsVI/Instances/BinaryReverseAdd.lean:486 Instances.BinaryReverseAdd.pre_entry_word_transitions (theorem) lean/SixBirdsFoundationsVI/Instances/BinaryReverseAdd.lean:520 Instances.BinaryReverseAdd.pre_entry_orbit (theorem) lean/SixBirdsFoundationsVI/Instances/BinaryReverseAdd.lean:533 Instances.BinaryReverseAdd.confined_after_entry (theorem) lean/SixBirdsFoundationsVI/Instances/BinaryReverseAdd.lean:559 Instances.BinaryReverseAdd.orbit_22_never_palindrome (theorem) lean/SixBirdsFoundationsVI/Instances/BinaryReverseAdd.lean:567
Corollary 4.7	G5, instance catalog line 749	Instances.BinaryReverseAdd.pre_entry_word_transitions (theorem) lean/SixBirdsFoundationsVI/Instances/BinaryReverseAdd.lean:520 Instances.BinaryReverseAdd.pre_entry_orbit (theorem) lean/SixBirdsFoundationsVI/Instances/BinaryReverseAdd.lean:533 Instances.BinaryReverseAdd.confined_after_entry (theorem) lean/SixBirdsFoundationsVI/Instances/BinaryReverseAdd.lean:559 Instances.BinaryReverseAdd.orbit_22_never_palindrome (theorem) lean/SixBirdsFoundationsVI/Instances/BinaryReverseAdd.lean:567
Theorem 4.18	G10, instance catalog line 1291	Instances.Ducci.iter_pow_two (theorem) lean/SixBirdsFoundationsVI/Instances/Ducci.lean:17 Instances.Ducci.ducci_nilpotent (theorem) lean/SixBirdsFoundationsVI/Instances/Ducci.lean:38 Instances.Ducci.ducci_vector_nilpotent (theorem) lean/SixBirdsFoundationsVI/Instances/Ducci.lean:80

Item	Law and part	Lean declarations
Remark 4.19	G10, theorem for $n = 4$; general n on paper catalog line 1291	<code>Instances.IntegerDucci4.step (def)</code> <code>lean/SixBirdsFoundationsVI/Instances/IntegerDucci4.lean:8</code> <code>Instances.IntegerDucci4.maximum_nonincreasing (theorem)</code> <code>lean/SixBirdsFoundationsVI/Instances/IntegerDucci4.lean:29</code> <code>Instances.IntegerDucci4.four_steps_even (theorem)</code> <code>lean/SixBirdsFoundationsVI/Instances/IntegerDucci4.lean:59</code> <code>Instances.IntegerDucci4.repeated_four_divisibility (theorem)</code> <code>lean/SixBirdsFoundationsVI/Instances/IntegerDucci4.lean:131</code> <code>Instances.IntegerDucci4.reaches_zero (theorem)</code> <code>lean/SixBirdsFoundationsVI/Instances/IntegerDucci4.lean:162</code>
Definition 4.9	G8, setting catalog line 1073	<code>Laws.G8OdometerAbelianization.ApplyRun (def)</code> <code>lean/SixBirdsFoundationsVI/Laws/G8OdometerAbelianization.lean:20</code> <code>Laws.G8OdometerAbelianization.LegalRun (inductive)</code> <code>lean/SixBirdsFoundationsVI/Laws/G8OdometerAbelianization.lean:33</code> <code>Laws.G8OdometerAbelianization.RunCounter (def)</code> <code>lean/SixBirdsFoundationsVI/Laws/G8OdometerAbelianization.lean:46</code> <code>Laws.G8OdometerAbelianization.Stable (def)</code> <code>lean/SixBirdsFoundationsVI/Laws/G8OdometerAbelianization.lean:64</code>
Definition 4.10	G8, hypothesis catalog line 1073	<code>Laws.G8OdometerAbelianization.AbelianCompatible (def)</code> <code>lean/SixBirdsFoundationsVI/Laws/G8OdometerAbelianization.lean:126</code>
Theorem 4.11	G8, Part A, with termination catalog line 1073	<code>Laws.G8OdometerAbelianization.partA_order_independent_and_odometer (theorem)</code> <code>lean/SixBirdsFoundationsVI/Laws/G8OdometerAbelianization.lean:248</code> <code>Laws.G8OdometerAbelianization.order_independent_stabilization (theorem)</code> <code>lean/SixBirdsFoundationsVI/Laws/G8OdometerAbelianization.lean:279</code> <code>Laws.G8OdometerAbelianization.odometer_invariance (theorem)</code> <code>lean/SixBirdsFoundationsVI/Laws/G8OdometerAbelianization.lean:295</code>
Example 4.13	G8, case (b) catalog line 1073	<code>Laws.G8OdometerAbelianization.caseB_not_abelian (theorem)</code> <code>lean/SixBirdsFoundationsVI/Laws/G8OdometerAbelianization.lean:493</code> <code>Laws.G8OdometerAbelianization.caseB_confluent_runs_share_final (theorem)</code> <code>lean/SixBirdsFoundationsVI/Laws/G8OdometerAbelianization.lean:737</code> <code>Laws.G8OdometerAbelianization.caseB_route_dependent_counter (theorem)</code> <code>lean/SixBirdsFoundationsVI/Laws/G8OdometerAbelianization.lean:767</code>
Example 4.14	G8, case (d) catalog line 1073	<code>Laws.G8OdometerAbelianization.caseD_final_states_distinct (theorem)</code> <code>lean/SixBirdsFoundationsVI/Laws/G8OdometerAbelianization.lean:877</code> <code>Laws.G8OdometerAbelianization.caseD_nonconfluent_witness (theorem)</code> <code>lean/SixBirdsFoundationsVI/Laws/G8OdometerAbelianization.lean:884</code>
Remark 4.15	G8, Part B catalog line 1073	<code>Laws.G8OdometerAbelianization.LeastActionMonotonicity (def)</code> <code>lean/SixBirdsFoundationsVI/Laws/G8OdometerAbelianization.lean:150</code> <code>Laws.G8OdometerAbelianization.least_action (theorem)</code> <code>lean/SixBirdsFoundationsVI/Laws/G8OdometerAbelianization.lean:414</code>
Proposition 6.6	G1, Part (a) catalog line 155	<code>Laws.G1HiddenAmortizedSolvency.PointwiseLiveness (def)</code> <code>lean/SixBirdsFoundationsVI/Laws/G1HiddenAmortizedSolvency.lean:47</code> <code>Laws.G1HiddenAmortizedSolvency.EmptyBadTailIntersection (def)</code> <code>lean/SixBirdsFoundationsVI/Laws/G1HiddenAmortizedSolvency.lean:56</code> <code>Laws.G1HiddenAmortizedSolvency.part_a_reduction (theorem)</code> <code>lean/SixBirdsFoundationsVI/Laws/G1HiddenAmortizedSolvency.lean:69</code>
Template 6.7	G1, template, Part (b) catalog line 155	<code>Laws.G1HiddenAmortizedSolvency.GhostConvergence (def)</code> <code>lean/SixBirdsFoundationsVI/Laws/G1HiddenAmortizedSolvency.lean:101</code> <code>Laws.G1HiddenAmortizedSolvency.NativeSeparation (def)</code> <code>lean/SixBirdsFoundationsVI/Laws/G1HiddenAmortizedSolvency.lean:112</code> <code>Laws.G1HiddenAmortizedSolvency.part_b_discharge (theorem)</code> <code>lean/SixBirdsFoundationsVI/Laws/G1HiddenAmortizedSolvency.lean:127</code>
Theorem 5.1	G2, theorem catalog line 303	<code>Laws.G2TransfiniteEscrow.no_infinite_nonterminal_run (theorem)</code> <code>lean/SixBirdsFoundationsVI/Laws/G2TransfiniteEscrow.lean:85</code> <code>Laws.G2TransfiniteEscrow.not_nonterminal_of_stage_coherence (theorem)</code> <code>lean/SixBirdsFoundationsVI/Laws/G2TransfiniteEscrow.lean:110</code> <code>Instances.G2NativeCoherence.NativeStageCoherence (def)</code> <code>lean/SixBirdsFoundationsVI/Instances/G2NativeCoherence.lean:8</code> <code>Instances.G2NativeCoherence.no_infinite_nonterminal_run_native (theorem)</code> <code>lean/SixBirdsFoundationsVI/Instances/G2NativeCoherence.lean:16</code>

Item	Law and part	Lean declarations
Theorem 4.1	G3, theorem catalog line 439	Laws.G3AmortizedCurrency.telescoping_identity (theorem) lean/SixBirdsFoundationsVI/Laws/G3AmortizedCurrency.lean:114 Laws.G3AmortizedCurrency.telescoping_bound (theorem) lean/SixBirdsFoundationsVI/Laws/G3AmortizedCurrency.lean:133 Laws.G3AmortizedCurrency.amortized_sum_le_scale (theorem) lean/SixBirdsFoundationsVI/Laws/G3AmortizedCurrency.lean:146 Laws.G3AmortizedCurrency.uniform_actual_cost_bound (theorem) lean/SixBirdsFoundationsVI/Laws/G3AmortizedCurrency.lean:172
Proposition S4.1 Theorem 4.3	G1–G3, corollary G4, general statement catalog line 610	argued on paper; not formalized Laws.G4MovingCoverExhaustion.positive_schema (theorem) lean/SixBirdsFoundationsVI/Laws/G4MovingCoverExhaustion.lean:41 Laws.G4MovingCoverExhaustion.conditional_biconditional (theorem) lean/SixBirdsFoundationsVI/Laws/G4MovingCoverExhaustion.lean:68 Laws.G4MovingCoverExhaustion.fixed_package_no_go (theorem) lean/SixBirdsFoundationsVI/Laws/G4MovingCoverExhaustion.lean:97
Example 4.2	G3, instance catalog line 439	Instances.BinaryCounter.increment (def) lean/SixBirdsFoundationsVI/Instances/BinaryCounter.lean:6 Instances.BinaryCounter.cost_is_bit_changes (theorem) lean/SixBirdsFoundationsVI/Instances/BinaryCounter.lean:27 Instances.BinaryCounter.potential_nonnegative (theorem) lean/SixBirdsFoundationsVI/Instances/BinaryCounter.lean:39 Instances.BinaryCounter.amortized_exact (theorem) lean/SixBirdsFoundationsVI/Instances/BinaryCounter.lean:49 Instances.BinaryCounter.n_increments_from_zero (theorem) lean/SixBirdsFoundationsVI/Instances/BinaryCounter.lean:86
Example 4.12	G8, instance on three sites; general graphs on paper catalog line 1073	Instances.TriangleSandpile.topple (def) lean/SixBirdsFoundationsVI/Instances/TriangleSandpile.lean:14 Instances.TriangleSandpile.h1 (theorem) lean/SixBirdsFoundationsVI/Instances/TriangleSandpile.lean:47 Instances.TriangleSandpile.chips_drop_one (theorem) lean/SixBirdsFoundationsVI/Instances/TriangleSandpile.lean:54 Instances.TriangleSandpile.h2 (theorem) lean/SixBirdsFoundationsVI/Instances/TriangleSandpile.lean:59 Instances.TriangleSandpile.stabilizes (theorem) lean/SixBirdsFoundationsVI/Instances/TriangleSandpile.lean:72 Instances.TriangleSandpile.canonical_stabilization (theorem) lean/SixBirdsFoundationsVI/Instances/TriangleSandpile.lean:91
Example 4.21	G12, instance, combinatorial part; embedding on paper catalog line 1483	Instances.MoserSpindle.edges (def) lean/SixBirdsFoundationsVI/Instances/MoserSpindle.lean:10 Instances.MoserSpindle.no_fin3_coloring (theorem) lean/SixBirdsFoundationsVI/Instances/MoserSpindle.lean:40 Instances.MoserSpindle.spindle_witness (theorem) lean/SixBirdsFoundationsVI/Instances/MoserSpindle.lean:46 Instances.MoserSpindle.no_coloring_of_edge_image (theorem) lean/SixBirdsFoundationsVI/Instances/MoserSpindle.lean:57
Example 4.4	G4, instance: exhaustion and leak discharged catalog line 610	Instances.MovingCover.Region (def) lean/SixBirdsFoundationsVI/Instances/MovingCover.lean:8 Instances.MovingCover.residual (def) lean/SixBirdsFoundationsVI/Instances/MovingCover.lean:11 Instances.MovingCover.residual_decreases (theorem) lean/SixBirdsFoundationsVI/Instances/MovingCover.lean:13 Instances.MovingCover.exhausted (theorem) lean/SixBirdsFoundationsVI/Instances/MovingCover.lean:26 Instances.MovingCover.finite_leak (theorem) lean/SixBirdsFoundationsVI/Instances/MovingCover.lean:42 Instances.MovingCover.no_finite_package (theorem) lean/SixBirdsFoundationsVI/Instances/MovingCover.lean:50
Template 4.8	G5, no-go template catalog line 749	Laws.G5CarryHorizonConfinement.UnboundedHorizon (def) lean/SixBirdsFoundationsVI/Laws/G5CarryHorizonConfinement.lean:32 Laws.G5CarryHorizonConfinement.no_fixed_depth_future_sufficient (theorem) lean/SixBirdsFoundationsVI/Laws/G5CarryHorizonConfinement.lean:42 Laws.G5CarryHorizonConfinement.no_future_sufficient_depth (theorem) lean/SixBirdsFoundationsVI/Laws/G5CarryHorizonConfinement.lean:53

Item	Law and part	Lean declarations
Theorem 4.5	G5, confinement catalog line 749	<code>Laws.G5CarryHorizonConfinement.closed_pattern_persists</code> (theorem) <code>lean/SixBirdsFoundationsVI/Laws/G5CarryHorizonConfinement.lean:118</code> <code>Laws.G5CarryHorizonConfinement.closed_pattern_confinement</code> (theorem) <code>lean/SixBirdsFoundationsVI/Laws/G5CarryHorizonConfinement.lean:138</code>
Proposition 6.8	G6, template: elementary final steps; rate bridge open catalog line 887	<code>Laws.G6EndogenousNeedleGeneration.DominantPressure</code> (def) <code>lean/SixBirdsFoundationsVI/Laws/G6EndogenousNeedleGeneration.lean:21</code> <code>Laws.G6EndogenousNeedleGeneration.covering_schema</code> (theorem) <code>lean/SixBirdsFoundationsVI/Laws/G6EndogenousNeedleGeneration.lean:32</code> <code>Laws.G6EndogenousNeedleGeneration.DominantObstruction</code> (def) <code>lean/SixBirdsFoundationsVI/Laws/G6EndogenousNeedleGeneration.lean:63</code> <code>Laws.G6EndogenousNeedleGeneration.hole_forming_schema</code> (theorem) <code>lean/SixBirdsFoundationsVI/Laws/G6EndogenousNeedleGeneration.lean:73</code>
Theorem 5.2	G7, certificate checks; part 3 needs a state in the invariant catalog line 979	<code>Laws.G7AdversarialMobilityConfinement.escape_never_stuck</code> (theorem) <code>lean/SixBirdsFoundationsVI/Laws/G7AdversarialMobilityConfinement.lean:71</code> <code>Laws.G7AdversarialMobilityConfinement.confinement_forces_trap</code> (theorem) <code>lean/SixBirdsFoundationsVI/Laws/G7AdversarialMobilityConfinement.lean:222</code> <code>Laws.G7AdversarialMobilityConfinement.escape_confinement_mutually_exclusive</code> (theorem) <code>lean/SixBirdsFoundationsVI/Laws/G7AdversarialMobilityConfinement.lean:269</code>
Theorem 6.9	G9, census bound catalog line 1202	<code>Laws.G9DefectEvacuationTransport.Census</code> (def) <code>lean/SixBirdsFoundationsVI/Laws/G9DefectEvacuationTransport.lean:21</code> <code>Laws.G9DefectEvacuationTransport.census_tail_bound</code> (theorem) <code>lean/SixBirdsFoundationsVI/Laws/G9DefectEvacuationTransport.lean:64</code> <code>Laws.G9DefectEvacuationTransport.census_excludes_unbounded_creation</code> (theorem) <code>lean/SixBirdsFoundationsVI/Laws/G9DefectEvacuationTransport.lean:80</code>
Template 6.10	G9, template; Lean has the packaging step only catalog line 1202	<code>Laws.G9DefectEvacuationTransport.TransportData</code> (structure) <code>lean/SixBirdsFoundationsVI/Laws/G9DefectEvacuationTransport.lean:97</code> <code>Laws.G9DefectEvacuationTransport.certified_transport_regime_of_assigned_nonzero</code> (theorem) <code>lean/SixBirdsFoundationsVI/Laws/G9DefectEvacuationTransport.lean:122</code>

Item	Law and part	Lean declarations
Proposition 6.11	G9, partial results for rule 184 catalog line 1202	Instances.Rule184.step (def) lean/SixBirdsFoundationsVI/Instances/Rule184.lean:10 Instances.Rule184.Jam (def) lean/SixBirdsFoundationsVI/Instances/Rule184.lean:93 Instances.Rule184.free_flow_shift (theorem) lean/SixBirdsFoundationsVI/Instances/Rule184.lean:151 Instances.Rule184.free_flow_invariant (theorem) lean/SixBirdsFoundationsVI/Instances/Rule184.lean:175 Instances.Rule184.free_flow_tail (theorem) lean/SixBirdsFoundationsVI/Instances/Rule184.lean:208 Instances.Rule184.step_jam (theorem) lean/SixBirdsFoundationsVI/Instances/Rule184.lean:270 Instances.Rule184.step_freeA (theorem) lean/SixBirdsFoundationsVI/Instances/Rule184.lean:274 Instances.Rule184.step_freeB (theorem) lean/SixBirdsFoundationsVI/Instances/Rule184.lean:278 Instances.Rule184.jam_has_one_blocked (theorem) lean/SixBirdsFoundationsVI/Instances/Rule184.lean:297 Instances.Rule184.orbit4_blocked_zero (theorem) lean/SixBirdsFoundationsVI/Instances/Rule184.lean:299 Instances.Rule184.plateau_low_density (theorem) lean/SixBirdsFoundationsVI/Instances/Rule184.lean:236 Instances.Rule184.plateau_has_jam (theorem) lean/SixBirdsFoundationsVI/Instances/Rule184.lean:237 Instances.Rule184.plateau_jam_count_stalls (theorem) lean/SixBirdsFoundationsVI/Instances/Rule184.lean:238 Instances.Rule184.low_density_four_one_step (theorem) lean/SixBirdsFoundationsVI/Instances/Rule184.lean:338 Instances.Rule184.Conservation.cars_conserved (theorem) lean/SixBirdsFoundationsVI/Instances/Rule184.Conservation.lean:86 Instances.Rule184Small.clears_up_to_eight (theorem) lean/SixBirdsFoundationsVI/Instances/Rule184Small.lean:252 Laws.G10LossfulPersistence.boundary_persistence (theorem) lean/SixBirdsFoundationsVI/Laws/G10LossfulPersistence.lean:57
Theorem 4.16	G10, general statement catalog line 1291	Laws.G10LossfulPersistence.checksumD (def) lean/SixBirdsFoundationsVI/Laws/G10LossfulPersistence.lean:94 Laws.G10LossfulPersistence.checksum_lossful (theorem) lean/SixBirdsFoundationsVI/Laws/G10LossfulPersistence.lean:119 Laws.G10LossfulPersistence.checksum_protect (theorem) lean/SixBirdsFoundationsVI/Laws/G10LossfulPersistence.lean:130 Laws.G10LossfulPersistence.checksum_regen (theorem) lean/SixBirdsFoundationsVI/Laws/G10LossfulPersistence.lean:142 Laws.G10LossfulPersistence.checksum_boundary_persistence (theorem) lean/SixBirdsFoundationsVI/Laws/G10LossfulPersistence.lean:155
Example 4.17	G10, instance: protection and regeneration discharged catalog line 1291	Laws.G10LossfulPersistence.checksumD (def) lean/SixBirdsFoundationsVI/Laws/G10LossfulPersistence.lean:94 Laws.G10LossfulPersistence.checksum_lossful (theorem) lean/SixBirdsFoundationsVI/Laws/G10LossfulPersistence.lean:119 Laws.G10LossfulPersistence.checksum_protect (theorem) lean/SixBirdsFoundationsVI/Laws/G10LossfulPersistence.lean:130 Laws.G10LossfulPersistence.checksum_regen (theorem) lean/SixBirdsFoundationsVI/Laws/G10LossfulPersistence.lean:142 Laws.G10LossfulPersistence.checksum_boundary_persistence (theorem) lean/SixBirdsFoundationsVI/Laws/G10LossfulPersistence.lean:155
Template 6.12	G11, template catalog line 1394	Laws.G11GlobalAntiSymmetry.Hierarchy (def) lean/SixBirdsFoundationsVI/Laws/G11GlobalAntiSymmetry.lean:66 Laws.G11GlobalAntiSymmetry.ForcedBreaksPeriod (def) lean/SixBirdsFoundationsVI/Laws/G11GlobalAntiSymmetry.lean:79 Laws.G11GlobalAntiSymmetry.hierarchy_forces_aperiodic (theorem) lean/SixBirdsFoundationsVI/Laws/G11GlobalAntiSymmetry.lean:103 Laws.G11GlobalAntiSymmetry.g11_global_anti_symmetry (theorem) lean/SixBirdsFoundationsVI/Laws/G11GlobalAntiSymmetry.lean:121 Laws.G12FiniteWitnessRadiation.no_global_k_coloring (theorem) lean/SixBirdsFoundationsVI/Laws/G12FiniteWitnessRadiation.lean:44 Laws.G12FiniteWitnessRadiation.orbit_radiation (theorem) lean/SixBirdsFoundationsVI/Laws/G12FiniteWitnessRadiation.lean:92 Laws.G12FiniteWitnessRadiation.CompactnessBridge (def) lean/SixBirdsFoundationsVI/Laws/G12FiniteWitnessRadiation.lean:117 Laws.G12FiniteWitnessRadiation.finite_witness_iff_no_global_coloring (theorem) lean/SixBirdsFoundationsVI/Laws/G12FiniteWitnessRadiation.lean:126
Theorem 4.20	G12, theorem catalog line 1483	Laws.G12FiniteWitnessRadiation.no_global_k_coloring (theorem) lean/SixBirdsFoundationsVI/Laws/G12FiniteWitnessRadiation.lean:44 Laws.G12FiniteWitnessRadiation.orbit_radiation (theorem) lean/SixBirdsFoundationsVI/Laws/G12FiniteWitnessRadiation.lean:92
Proposition 4.22	G12, bridge catalog line 1483	Laws.G12FiniteWitnessRadiation.CompactnessBridge (def) lean/SixBirdsFoundationsVI/Laws/G12FiniteWitnessRadiation.lean:117 Laws.G12FiniteWitnessRadiation.finite_witness_iff_no_global_coloring (theorem) lean/SixBirdsFoundationsVI/Laws/G12FiniteWitnessRadiation.lean:126
Theorem 4.23	G13, conditional on its count hypotheses catalog line 1603	Laws.G13ThinOrbitSaturation.relative_density_theorem (theorem) lean/SixBirdsFoundationsVI/Laws/G13ThinOrbitSaturation.lean:54

Item	Law and part	Lean declarations
Example 4.24	G13, weak instance: powers of two catalog line 1603	<code>Instances.ThinOrbit.powers</code> (def) <code>lean/SixBirdsFoundationsVI/Instances/ThinOrbit.lean:109</code> <code>Instances.ThinOrbit.power_members_iff</code> (theorem) <code>lean/SixBirdsFoundationsVI/Instances/ThinOrbit.lean:85</code> <code>Instances.ThinOrbit.power_count_sublinear</code> (theorem) <code>lean/SixBirdsFoundationsVI/Instances/ThinOrbit.lean:143</code> <code>Instances.ThinOrbit.powers_infinite</code> (theorem) <code>lean/SixBirdsFoundationsVI/Instances/ThinOrbit.lean:160</code> <code>Instances.ThinOrbit.powers_relative_zero</code> (theorem) <code>lean/SixBirdsFoundationsVI/Instances/ThinOrbit.lean:166</code> <code>Instances.ThinOrbit.counted_exceptions</code> (theorem) <code>lean/SixBirdsFoundationsVI/Instances/ThinOrbit.lean:19</code> <code>Instances.ThinOrbit.concrete_zero_density_failure</code> (theorem) <code>lean/SixBirdsFoundationsVI/Instances/ThinOrbit.lean:172</code>
Template 4.25	G13, template; via the density theorem, conditional on its count hypotheses and the imported bound catalog line 1603	<code>Laws.G13ThinOrbitSaturation.relative_density_theorem</code> (theorem) <code>lean/SixBirdsFoundationsVI/Laws/G13ThinOrbitSaturation.lean:54</code>

Table S2. Catalog status and the status used in the main paper.

Law	Catalog name	Catalog status	Main paper	Line
G1	Hidden Amortized Solvency	theorem/schema split	template (with a reformulation)	46
G2	Transfinite Escrow	theorem	general theorem	264
G3	Amortized Potential Currency	theorem	law	405
G4	Moving-Cover Exhaustion	schema	law	577
G5	Carry-Horizon Confinement	calibration-anchored schema	law (no-go part a template)	714
G6	Endogenous Needle Generation	schema	template	836
G7	Adversarial Mobility Confinement	calibration-anchored schema	general theorem	953
G8	Odometer Abelianization	theorem	law	1039
G9	Defect-Evacuation-to- Transport	calibration-anchored schema	template	1154
G10	Lossful Boundary Persistence	theorem	law	1257
G11	Local-Rule Global-Anti-Symmetry	theorem	template	1372
G12	Finite Witness Radiation	theorem	law	1443
G13	Thin-Orbit Saturation	schema	law, weak instance (saturation part a template)	1542

S3 Case analyses and instances

S3.1 Case analyses

For each law, the table lists the principal cases a system can fall into and what settles a system in each case. Only the cases for G1 are exhaustive; for the other laws a system may fall outside every listed case, and then nothing in this paper applies to it. A case in which nothing has been proved is recorded as undetermined, never as a failure of the law.

Table S3. Case analyses.

Law	Case	Settled by
G1	(a) terminal, or descends at step 1	membership in A , or the one-step ledger inequality
	(b) first descends at a step $k > 1$	the ledger inequality at step k ; by Remark 6.5, k is unbounded for Collatz
	(c) never descends	a point in every N_k ; for Collatz, this is the open case
G2	(a) escrow in $\mathbb{N}$	a ranking function
	(b) escrow in a finite lexicographic product	a lexicographic ranking
	(c) escrow in a larger ordinal	a stage-coherent ordinal assignment (Goodstein)
	(d) no escrow of a declared strength	an independence result (Kirby–Paris for Peano arithmetic)
G3	(a) potential never increases	the telescoping bound
	(b) potential increases at some steps, amortized cost bounded	Theorem 4.1 (binary counter, dynamic array)
	(c) no potential from a declared class bounds the cost	a lower bound for the class
G4	(a) a sound exhaustive cover exists	Theorem 4.3 (initial intervals; greedy unit fractions)
	(b) a fixed finite package covers	a finite list of patches
	(c) a leak for the restricted class	(H4) of Theorem 4.3 (Mordell’s quadratic-residue obstruction)
	(d) undetermined	open (Erdős–Straus)
G5	(a) confined by a closed target-free family	Theorem 4.5 (orbit of 22)
	(b) reaches a target	a finite computation
	(c) no fixed depth predicts	split pairs at every depth, Template 4.8
	(d) undetermined	open (196 in base 10)
G6	(a) covering	a hole count that always later decreases (unit-jump variant)
	(b) permanent hole	non-visitation that propagates (jumps $2n$)
	(c) undetermined	open (Recamán’s sequence)
G7	(a) escape	an escape strategy with invariant (angel of power 2)
	(b) confinement	a rank-decreasing devil strategy
	(c) undetermined	neither certificate supplied
G8	(a) abelian, no infinite legal run	Theorem 4.11; odometer

Law	Case	Settled by
	(b) one final configuration, counters depend on the route	Example 4.13
	(c) some legal run is infinite	an infinite run; no odometer
	(d) terminating, several final configurations	Example 4.14
	(e) terminating, not abelian, one final configuration and the same counters on all maximal runs	direct computation; for example, maximal runs rst and srt from S with $m_s m_r(S) \neq m_r m_s(S)$, both sent by t to one stable configuration
G9	(a) all defects eventually resolved (b) certified transport	a census with empty persistent set transporter records with checked recurrences (rule 184)
	(c) defects persist without motion (d) unbounded unresolved creation (e) undetermined	stationary composites excluded by Theorem 6.9 under a census a finite census only
G10	(a) persistence proved (b) collapse (c) persistence seen up to a finite depth	protection, regeneration, initial invariant (checksum) a nilpotency certificate (Ducci, length 2^k) a computation; no claim beyond it (Gilbreath)
G11	(a) a periodic admissible configuration (b) admissible configurations exist, all aperiodic (c) no admissible configuration	a period and a fundamental domain existence plus Template 6.12 (Robinson) a finite region that cannot be filled no algorithm decides among (a)–(c) [8]
G12	(a) a valid finite witness (b) the proposed witness fails (c) the claim depends on choice	Theorem 4.20 (Moser spindle; de Grey’s graph) a colouring, a wrong edge, or unrealizable coordinates Proposition 4.22, with the axiom recorded
G13	(a) finitely many exceptions (b) density one with exceptions (c) positive density only	an imported finite-exception theorem Template 4.25 (Bourgain–Kontorovich) Bourgain–Fuchs exceptions are further sorted into congruence failures, reciprocity obstructions [62] and uncertified cases

S3.2 Instances

Table S4 lists five or six systems for each entry. The grades are those of the main paper: *checked* (proved in the main paper and in Lean), *written* (proved in the main paper only), *imported* (a cited theorem), *open* (the question is unsolved), and *analogy*: the system has the shape of the law, but its hypotheses have not been checked for a precise model, and nothing is claimed about it.

Table S4. Instances by law.

Law	System	Grade	Sources
G1	Collatz: ledger and split pairs; liveness	checked; open	Lagarias [75, 76], Tao [111]

Law	System	Grade	Sources
	Binary-counter increments	written	Tarjan [112], Cormen et al. [29]
	Size-change termination	imported	Lee et al. [80]
	Aliquot sequences	open	Guy and Selfridge [61], Guy [60]
	Switched and probabilistic termination	analogy	Liberzon [81], Agrawal et al. [1]
G2	Goodstein sequences and hydra games	written; imported	Goodstein [56], Kirby and Paris [71]
	Ranking functions	imported	Floyd [48]
	Termination by path orderings	imported	Dershowitz [36]
	Ordinal analysis	analogy	Gentzen [54]
	Multiset orderings; cascade termination	imported	Dershowitz and Manna [37], Tsiokos [114]
G3	Binary counter	written	Example 4.2; Tarjan [112], Cormen et al. [29]
	Doubling array	written	Cormen et al. [29]
	Splay trees	imported	Sleator and Tarjan [105]
	Storage functions of passive systems	analogy	Willems [120], Tsiokos [115]
	Energy-harvesting budgets	analogy	Ozel et al. [93], Yang and Ulukus [122]
G4	Initial intervals of $\mathbb{N}$	checked	Example 4.4
	Greedy unit fractions	written	Fibonacci [45], Sylvester [110], Eppstein [43]
	Erdős–Straus	open	Mordell [86], Elsholtz and Tao [42]
	Covering systems	imported	Hough [68]
	Symbolic integration	analogy	Risch [99], Bronstein [20]
	Counterexample-guided abstraction refinement	analogy	Clarke et al. [27]
G5	Reverse-and-add in base 2, orbit of 22	checked	OEIS Foundation Inc. [92], Brown [21]
	Lychrel candidates in base 10	open	Walker [119]
	Carry-dependent timing	analogy	Kocher [73]
	Rounding-error propagation	analogy	Higham [65]
	Automatic sequences	analogy	Allouche and Shallit [3]
G6	Recamán’s sequence	open; proposed audit	OEIS Foundation Inc. [90, 91]
	Two designed variants	written	Section 6.2
	Self-avoiding walks	analogy	Madras and Slade [84], Flory [47]
	Novelty-seeking exploration	analogy	Bellemare et al. [7]
	Online algorithms with irrevocable choices	analogy	Borodin and El-Yaniv [10], Boyar et al. [16]
G7	Angel game	imported	Berlekamp et al. [9], Conway [28], Máthé [85], Kloster [72], Bowditch [15], Gács [51]

Law	System	Grade	Sources
	Cops and robbers	analogy	Nowakowski and Winkler [88], Aigner and Fromme [2]
	Routing under adversarial failures	analogy	Borodin et al. [11]
	Navigation in supercritical percolation	analogy	Grimmett [58], Kesten [70]
	Motion planning among moving obstacles	analogy	LaValle [79], Fraichard and Asama [49]
G8	Sandpiles	written	Example 4.12
	Rotor-routing and the least-action principle	imported	Dhar [38], Fey et al. [44], Holroyd et al. [66]
	Confluence with counters in rewriting	analogy	Tsiokos [117]
	Load balancing and gossip averaging	analogy	Cybenko [31], Boyd et al. [17]
	Neuronal avalanches	analogy	Beggs and Plenz [6]
	Clearing in payment systems	analogy	Eisenberg and Noe [41]
G9	Rule 184	imported; partial results checked	Fuk�s [50], Nagel [87], Chowdhury et al. [26]
	Langton’s ant	open	Langton [78], Bunimovich and Troubetzkoy [22]
	Excitable media	analogy	Greenberg and Hastings [57], Durrett and Griffeath [40]
	Gliders in cellular automata	analogy	Gardner [52], Pivato [96]
	Traffic shock waves	analogy	Lighthill and Whitham [82], Richards [98], Daganzo [32]
G10	Checksum cascade	checked	Fletcher [46], Stone et al. [108]
	Ducci maps of length 2^k	checked ($\mathbb{F}_2$); written (integers)	Chamberland and Thomas [24], Breuer [18], Breuer et al. [19]
	Gilbreath’s conjecture	open	Odlyzko [89], Chase [25], Plouffe [97]
	Boundary conditions of difference schemes	analogy	Strikwerda [109], Gustafsson et al. [59], Coulombel and Faye [30]
	Edges under smoothing	analogy	Canny [23], Witkin [121]
G11	Aperiodic tile sets	imported	Berger [8], Robinson [101], Penrose [95], Smith et al. [106, 107]
	Quasicrystals	analogy	Shechtman et al. [104], Senechal [103]
	Subshifts of finite type	imported	Lind and Marcus [83], Durand et al. [39]
	Constraint satisfaction	analogy	Dechter [35]
	Synchronization strings	analogy	Haeupler and Shahrasbi [63]
G12	Moser spindle, $\chi(\mathbb{R}^2) \geq 4$	written	Example 4.21
	$\chi(\mathbb{R}^2) \geq 5$	imported	de Grey [34], Heule [64], Parts [94], de Bruijn and Erd�s [33]
	Hardness gadgets	analogy	Garey et al. [53]

Law	System	Grade	Sources
	Rigidity	analogy	Laman [77], Asimow and Roth [4]
	Excluded minors	analogy	Robertson and Seymour [100]
	Impossibility theorems in social choice	analogy	Gibbard [55], Satterthwaite [102]
G13	Powers of two (arithmetic example)	checked	Example 4.24
	Apollonian packings	imported	Bourgain and Fuchs [12], Bourgain and Kontorovich [13], Haag et al. [62]
	Zaremba's conjecture	imported (density one); open	Bourgain and Kontorovich [14]
	Affine sieve	analogy	Kontorovich and Oh [74]
	Expander graphs	analogy	Hoory et al. [67]
	Diffraction of model sets	analogy	Baake and Grimm [5]

S4 Three entries read as asymptotic statuses

The paper on audited operational realisability names four statuses of a nonnegative sequence $c_0, c_1, \dots$ [114]: *budgeted* ($c_j \leq B$ for all large j), *summable* ($\sum_j c_j < \infty$), *contractive* (on the nonzero tail the ratios c_{j+1}/c_j tend to a constant below 1) and *eventually zero*. Each implies the one before it. Only these facts about sequences are used here; nothing about that paper's other constructions is transferred.

Proposition S4.1 (The three currencies as statuses). *The following hold.*

1. *In the setting of G1 (Section 6.1), a point $x \notin A$ descends at some step exactly when the indicator of $x \in N_j$ is eventually zero.*
2. *Under the hypotheses of Theorem 5.1, the indicator that no stage up to k is terminal is eventually zero.*
3. *In the setting of G3 (Section 4.1), write $H = \sum_{i < J} \hat{a}_i$ for a head of the amortized costs. If the amortized costs are budgeted by B from J on, then $\sum_{i \leq n} a_i \leq \Phi(s_0) + H + B(n - J + 1)$ for $n \geq J$. If they are summable with sum S , then every partial sum of actual costs is at most $S + \Phi(s_0)$. If they are contractive, choose $q < 1$ and J with $\hat{a}_{i+1} \leq q \hat{a}_i$ for all $i \geq J$; then the same holds with S replaced by $H + \hat{a}_J/(1 - q)$. If they vanish from J on, every partial sum is at most $\Phi(s_0) + H$.*

Proof. Part 1 restates Proposition 6.6 for one point, since the sets N_j decrease. Part 2 follows from Theorem 5.1, with the terminal stage obtained classically. Part 3 combines Theorem 4.1 with the standard bound matching each status. $\square$

Origin. Elementary; argued on paper. The budgeted and eventually-zero cases of part 3 are readings of the integer statement checked in Lean; the summable and contractive cases need non-integer costs and are not formalized.

S5 Non-implications

Table S5 lists, law by law, statements that do not follow from what the main paper proves, with the example or reason that separates them. It extends Table 4 of the main paper.

Table S5. Non-implications.

Law	Does not follow	Separating example or reason
G1	Liveness of the Collatz map	Both hypotheses of Template 6.7 are unproved for it.
	A bound on the first descent time uniform in n	$2^L - 1$ stays above its start for $L - 1$ steps (Remark 6.5).
	Prediction of $a(n)$ from $n \bmod 2^m$	Theorem 6.3.
	Pointwise liveness from almost-all results	Tao’s theorem [111] concerns almost all starts and bounds values, not descent of every orbit.
	A statement about integers from one about the ghost	The ghost -1 is a 2-adic point, not an integer orbit; only integer inequalities are certificates.
G2	Termination provable in a given theory	Goodstein termination is not provable in Peano arithmetic [71].
	Descent that ignores the change of presentation	In Goodstein sequences the integer value grows while the ordinal falls; comparing values in different presentations proves nothing.
G3	Eventual success	Theorem 4.1 bounds cost over every finite stretch and says nothing about termination.
	A potential from a correlated statistic	Only a nonnegative potential in the telescoping identity gives the bound.
G4	Exhaustion from covering every checked target	Exhaustion is a hypothesis of Theorem 4.3; Erdős–Straus is verified far beyond any bound without a proof.
	Pointwise closure from density results	The results of Vaughan [118], Elsholtz and Tao [42] bound the exceptions without excluding them.
	A fixed package for Erdős–Straus	No finite family of polynomial identities covers the classes that are squares modulo every prime [86].
G5	Escape to non-palindromes without a closed family	The confinement theorem needs a closed target-free family; none is known for 196 in base 10.
	The unbounded-horizon hypothesis for base 2	Not proved here; the confinement of 22 does not use it.
G6	Coverage or a permanent hole from rate conditions	The bridge from rate conditions to the premises of Proposition 6.8 is open.
	A permanent hole from a long computation	852655 is missing from every computed prefix; this is not a proof.
	Either half of G6 for Recamán’s sequence	The premises of Proposition 6.8 have not been established for it.
	The system certifying itself	The obstruction is produced by the run and audited from outside.
G7	Winning strategies from Theorem 5.2	The theorem checks given strategies.
	Exclusion of the two certificates with an empty invariant	An empty set satisfies the escape hypothesis vacuously; part 3 needs a state in it.

Law	Does not follow	Separating example or reason
	Infinite-board results from finite tori Thresholds independent of the rules	A finite torus changes the game. The angel results depend on the metric, the power and the number of deletions per turn.
G8	Route-independent counters from confluence The Lean statement without termination Uniqueness of final configurations from termination The least-action principle from Theorem 4.11 An arrow of time or entropy production	Example 4.13. The Lean theorem assumes that no infinite legal run starts at x . Example 4.14. It compares with scripts that need not be legal runs; it is imported [44]. Nothing in G8 or G9 concerns thermodynamics.
G9	Evacuation for rule 184 at every low density The evacuation template in Lean Transport from visible motion The highway of Langton's ant Transport from the census bound alone	Not proved; the partial results are Proposition 6.11. Lean only repackages a supplied certified transporter with one assigned defect. Only the recurrence equality certifies it. Open; only unboundedness is proved [22]. Theorem 6.9 excludes unbounded creation; it produces no transporter.
G10	Recovery of lost information Persistence from observation Persistence resting on parity for length 2^k	The theorem preserves the boundary relation; the interior is lost (checksum). A finite run gives bounded evidence (Gilbreath). Theorem 4.18 destroys parity.
G11	Aperiodicity from local rules alone An algorithm classifying tile sets Existence of an admissible configuration That a material is an instance	A hierarchy must be supplied. Undecidable [8]. A separate input to Template 6.12. A precise local-rule model is required first.
G12	Completeness of finite witnesses without compactness The exact chromatic number of the plane A proof-checked certificate for every witness	Proposition 4.22 assumes it. Open; $5 \leq \chi(\mathbb{R}^2) \leq 7$. The 826-vertex graph of Section S6 has none.
G13	A thin-orbit result from Example 4.24 Local-to-global from congruence conditions Density one from positive density "Thin" or "thick" in any informal sense	The example is arithmetic; it involves no group, orbit or spectral gap. False for many packings [62]. Bourgain and Fuchs [12] alone gives positive density. Both are technical: infinite index and Zariski dense; positive density.
All	Progress on any open problem named here Novelty of imported results	None is claimed. They are cited with attribution.

Law	Does not follow	Separating example or reason
	Transfer of constructions from the paper on audited operational realisability	Only four statuses of sequences are used (Proposition S4.1).

S6 Finite computations

The computations below are exact (integer, rational or algebraic arithmetic, or a SAT solver) and finite. None of them is used in a proof in the main paper; each illustrates a law on a bounded range, and a bounded run is evidence only about that range. The parameters of every run are stored with its output; Section S9 gives the commands.

Table S6. Finite computations, by law.

Run	Computation	Outcome
G1-L1	Accelerated Collatz orbits of all 500000 odd $n < 10^6$, with the ledger identity of Theorem 6.2 checked at 2241405 checkpoints.	Every start other than 1 descends; the latest first descent is at step 111, for $n = 626331$.
G1-L2	All odd starts below 10^7 ; the $2^L - 1$ formula of Remark 6.5 for $L = 2, \dots, 80$.	Latest first descent at step 155, for $n = 8088063$. The formula holds for every L tested. A stronger guess, that the first descent of $2^L - 1$ occurs exactly at step $L - 1$, failed in all 79 cases (for $L = 2$, $n = 3$ descends at step 2); the main paper claims only “at least $L - 1$ ”.
G1-L3	A finite analogue of the ghost: the accelerated map on the 32768 odd residues modulo 2^{16} .	Apart from the fixed residue 1, every residue descends except 14563 and 21845, which never descend and reach the ghost of this finite model, the odd solution $21845 = (2^{16} - 1)/3$ of $3r + 1 \equiv 0 \pmod{2^{16}}$. In this finite model both hypotheses of Template 6.7 hold; this says nothing about the integers.
G1-L4	Aliquot sequences from 100000 starts with a cap of 10^{12} .	82728 reach 0, 2085 enter one of 22 cycles, and 15187 exceed the cap.
G2-L1	Goodstein sequences from $1, \dots, 30$ for 4 steps, checking that rewriting preserves the ordinal and subtraction lowers it.	All 116 steps satisfy both identities; the largest value reached has 120605 bits. Starts 1 and 2 terminate within the budget.
G3-L1	1000 binary-counter increments from 44142; 1000 dynamic-array insertions.	Amortized cost exactly 2 per increment; at most 3 per insertion, with equality at each of the 8 resizes.
G4-L1	Greedy unit-fraction expansions of all 124750 fractions p/q with $0 < p < q \leq 500$.	Every expansion terminates with strictly decreasing numerators.
G5-L1	The base-2 reverse-and-add orbit of 22 for 84 steps.	Every phase transition of Theorem 4.6 holds; the largest word has 48 bits.
G6-L1	10^7 steps of Recamán’s sequence.	The backward move is legal at 4999986 steps; 852655 is still missing.
G6-L2	The two designed variants of Section 6.2 for 10^6 steps.	The variant with jumps $2n$ visits 736749 even values and no odd value; the unit-jump variant visits exactly $\{0, \dots, n\}$ by step n .

Run	Computation	Outcome
G7-L1	Bounded-horizon minimax for the angel game on small tori, 8 parameter settings.	On the 4×4 torus with $b = 2$, power 1 is trapped and power 2 survives to the horizon; the 3×3 settings cannot distinguish powers 1 and 2, and $b = 1$ on 4×4 was inconclusive. A finite torus says nothing about the infinite board.
G8-L1	500 stabilizations of random sandpiles on square grids of sides 10 to 50, 100 random legal orders each.	Every order gives the same final configuration and odometer.
G8-L2	1000 random legal orders in Example 4.13.	Every run ends at N ; exactly three counter vectors occur.
G9-L1	Rule 184 on a ring of 2000 cells at densities $\frac{3}{10}, \frac{2}{5}, \frac{1}{2}, \frac{3}{5}, \frac{7}{10}$.	Each run reaches the predicted regime; the recurrence certificate holds on 128 consecutive windows, with period 1 and displacement $+1$ below density $\frac{1}{2}$ and -1 above.
G10-L1	Integer Ducci tuples of lengths 3 to 16.	Lengths 4, 8, 16 reach zero (at steps 4, 18, 98 in the recorded runs), and $(I + S)^k = 0$ over $\mathbb{F}_2$ exactly for these lengths; the tuples of the other lengths enter nonzero cycles.
G11-L1	SAT encodings of periodic tilings on tori.	A one-colour tile set tiles the 1×1 torus; the 11-tile set of Jeandel and Rao [69] tiles no $p_1 \times p_2$ torus with $p_1, p_2 \leq 4$.
G12-L1	An 826-vertex unit-distance graph from the Polymath16 project, in exact algebraic arithmetic.	All 4273 edges have unit length and 2000 sampled non-edges do not; the graph is not 4-colourable (SAT, 3304 variables, 22874 clauses). No independently checkable proof of unsatisfiability is available for this graph.
G13-L1	Apollonian packings from $(-1, 2, 2, 3)$ and $(-6, 11, 14, 15)$ up to curvature 300000.	Descartes' relation holds for every quadruple, and all curvatures lie in the expected residue classes modulo 24. The second packing contains no curvature of the form $2n^2, 3n^2$ or $6n^2$ in range, as the reciprocity obstruction predicts.

S7 Formal status

S7.1 The library

The Lean 4 library has thirteen modules for the entries G1–G13 and seven for the instances. It imports no external library, contains no `sorry` and declares no axioms of its own.

S7.2 Axioms

Table S7 gives, for every theorem named in Table S1, the axioms reported by `#print axioms`. The records for the law modules are dated 2 September 2026. The records for the instance modules are dated 26 September 2026. Only the three standard axioms of Lean's kernel occur.

Table S7. Axioms of the theorems in the concordance.

Declaration	Axioms	Record
<code>Instances.Collatz.accelerated_step</code>	<code>propext</code> , <code>Quot.sound</code>	instances report
<code>Instances.Collatz.affine_ledger</code>	<code>propext</code> , <code>Quot.sound</code>	instances report
<code>Instances.Collatz.accelerated_affine_ledger</code>	<code>propext</code> , <code>Quot.sound</code>	instances report

Declaration	Axioms	Record
Instances.Collatz.c_one_or_two	propext, Quot.sound	instances report
Instances.Collatz.first_formula	propext, Quot.sound	instances report
Instances.Collatz.split_pair	propext, Quot.sound	instances report
Instances.Collatz.split_pair_next_exponent	propext, Quot.sound	instances report
Instances.Collatz.split_pair_prior_exponent	propext, Quot.sound	instances report
Instances.BinaryReverseAdd.transition_01	propext, Quot.sound	instances report
Instances.BinaryReverseAdd.transition_12	propext, Quot.sound	instances report
Instances.BinaryReverseAdd.transition_23	propext, Quot.sound	instances report
Instances.BinaryReverseAdd.transition_30	propext, Quot.sound	instances report
Instances.BinaryReverseAdd.word_transition_01	propext, Classical.choice, Quot.sound	instances report
Instances.BinaryReverseAdd.word_transition_12	propext, Classical.choice, Quot.sound	instances report
Instances.BinaryReverseAdd.word_transition_23	propext, Classical.choice, Quot.sound	instances report
Instances.BinaryReverseAdd.word_transition_30	propext, Classical.choice, Quot.sound	instances report
Instances.BinaryReverseAdd.p0_not_palindrome	propext, Quot.sound	instances report
Instances.BinaryReverseAdd.p1_not_palindrome	propext	instances report
Instances.BinaryReverseAdd.p2_not_palindrome	propext	instances report
Instances.BinaryReverseAdd.p3_not_palindrome	propext	instances report
Instances.BinaryReverseAdd.family_closed	propext, Classical.choice, Quot.sound	instances report
Instances.BinaryReverseAdd.family_target_free	propext, Classical.choice, Quot.sound	instances report
Instances.BinaryReverseAdd.pre_entry_word_transitions	propext, Classical.choice, Quot.sound	instances report
Instances.BinaryReverseAdd.pre_entry_orbit	propext, Classical.choice, Quot.sound	instances report
Instances.BinaryReverseAdd.confined_after_entry	propext, Classical.choice, Quot.sound	instances report
Instances.BinaryReverseAdd.orbit_22_never_palindrome	propext, Classical.choice, Quot.sound	instances report
Instances.Ducci.iter_pow_two	propext, Quot.sound	instances report
Instances.Ducci.ducci_nilpotent	propext, Quot.sound	instances report
Instances.Ducci.ducci_vector_nilpotent	propext, Quot.sound	instances report
Instances.IntegerDucci4.maximum_nonincreasing	propext, Quot.sound	2026-09-26 transcript
Instances.IntegerDucci4.four_steps_even	propext, Quot.sound	2026-09-26 transcript
Instances.IntegerDucci4.repeated_four_divisibility	propext, Quot.sound	2026-09-26 transcript
Instances.IntegerDucci4.reaches_zero	propext, Quot.sound	2026-09-26 transcript
Laws.G80dometerAbelianization.partA_order_independent_and_odometer	propext, Quot.sound	2026-09-02 receipt
Laws.G80dometerAbelianization.order_independent_stabilization	propext, Quot.sound	2026-09-02 receipt
Laws.G80dometerAbelianization.odometer_invariance	propext, Quot.sound	2026-09-02 receipt
Laws.G80dometerAbelianization.caseB_not_abelian	propext	2026-09-02 receipt
Laws.G80dometerAbelianization.caseB_confluent_runs_share_final	propext	2026-09-02 receipt
Laws.G80dometerAbelianization.caseB_route_dependent_counter	none	2026-09-02 receipt
Laws.G80dometerAbelianization.caseD_final_states_distinct	none	2026-09-02 receipt
Laws.G80dometerAbelianization.caseD_nonconfluent_witness	propext	2026-09-02 receipt
Laws.G80dometerAbelianization.least_action	propext, Quot.sound	2026-09-02 receipt
Laws.G1HiddenAmortizedSolvency.part_a_reduction	propext, Classical.choice, Quot.sound	2026-09-02 receipt
Laws.G1HiddenAmortizedSolvency.part_b_discharge	propext, Classical.choice, Quot.sound	2026-09-02 receipt
Laws.G2TransfiniteEscrow.no_infinite_nonterminal_run	none	2026-09-02 receipt
Laws.G2TransfiniteEscrow.not_nonterminal_of_stage_coherence	none	2026-09-02 receipt
Instances.G2NativeCoherence.no_infinite_nonterminal_run_native	none	instances report
Laws.G3AmortizedCurrency.telescoping_identity	propext, Quot.sound	2026-09-02 receipt
Laws.G3AmortizedCurrency.telescoping_bound	propext, Quot.sound	2026-09-02 receipt
Laws.G3AmortizedCurrency.amortized_sum_le_scale	propext, Quot.sound	2026-09-02 receipt
Laws.G3AmortizedCurrency.uniform_actual_cost_bound	propext, Quot.sound	2026-09-02 receipt

Declaration	Axioms	Record
Laws.G4MovingCoverExhaustion.positive_schema	none	2026-09-02 receipt
Laws.G4MovingCoverExhaustion.conditional_biconditional	none	2026-09-02 receipt
Laws.G4MovingCoverExhaustion.fixed_package_no_go	none	2026-09-02 receipt
Instances.BinaryCounter.cost_is_bit_changes	propext	2026-09-26 transcript
Instances.BinaryCounter.potential_nonnegative	propext, Quot.sound	2026-09-26 transcript
Instances.BinaryCounter.amortized_exact	propext, Quot.sound	2026-09-26 transcript
Instances.BinaryCounter.n_increments_from_zero	propext, Quot.sound	2026-09-26 transcript
Instances.TriangleSandpile.h1	propext, Quot.sound	2026-09-26 transcript
Instances.TriangleSandpile.chips_drop_one	propext, Quot.sound	2026-09-26 transcript
Instances.TriangleSandpile.h2	propext, Quot.sound	2026-09-26 transcript
Instances.TriangleSandpile.stabilizes	propext, Classical.choice, Quot.sound	2026-09-26 transcript
Instances.TriangleSandpile.canonical_stabilization	propext, Classical.choice, Quot.sound	2026-09-26 transcript
Instances.MoserSpindle.no_fin3_coloring	propext, Quot.sound	2026-09-26 transcript
Instances.MoserSpindle.spindle_witness	propext, Quot.sound	2026-09-26 transcript
Instances.MoserSpindle.no_coloring_of_edge_image	propext, Quot.sound	2026-09-26 transcript
Instances.MovingCover.residual_decreases	propext, Quot.sound	2026-09-26 transcript
Instances.MovingCover.exhausted	propext, Quot.sound	2026-09-26 transcript
Instances.MovingCover.finite_leak	propext, Quot.sound	2026-09-26 transcript
Instances.MovingCover.no_finite_package	propext, Quot.sound	2026-09-26 transcript
Laws.G5CarryHorizonConfinement.no_fixed_depth_future_sufficient	none	2026-09-02 receipt
Laws.G5CarryHorizonConfinement.no_future_sufficient_depth	none	2026-09-02 receipt
Laws.G5CarryHorizonConfinement.closed_pattern_persists	none	2026-09-02 receipt
Laws.G5CarryHorizonConfinement.closed_pattern_confinement	propext	2026-09-02 receipt
Laws.G6EndogenousNeedleGeneration.covering_schema	none	2026-09-02 receipt
Laws.G6EndogenousNeedleGeneration.hole_forming_schema	none	2026-09-02 receipt
Laws.G7AdversarialMobilityConfinement.escape_never_stuck	none	2026-09-02 receipt
Laws.G7AdversarialMobilityConfinement.confinement_forces_trap	propext, Classical.choice, Quot.sound	2026-09-02 receipt
Laws.G7AdversarialMobilityConfinement.escape_confinement_mutually_exclusive	propext	2026-09-02 receipt
Laws.G9DefectEvacuationTransport.census_tail_bound	propext	2026-09-02 receipt
Laws.G9DefectEvacuationTransport.census_excludes_unbound_creation	propext	2026-09-02 receipt
Laws.G9DefectEvacuationTransport.certified_transport_regime_of_assigned_nonzero	none	2026-09-02 receipt
Instances.Rule184.free_flow_shift	propext, Quot.sound	2026-09-26 transcript
Instances.Rule184.free_flow_invariant	propext, Quot.sound	2026-09-26 transcript
Instances.Rule184.free_flow_tail	propext, Quot.sound	2026-09-26 transcript
Instances.Rule184.step_jam	propext, Quot.sound	2026-09-26 transcript
Instances.Rule184.step_freeA	propext, Quot.sound	2026-09-26 transcript
Instances.Rule184.step_freeB	propext, Quot.sound	2026-09-26 transcript
Instances.Rule184.jam_has_one_blocked	propext	2026-09-26 transcript
Instances.Rule184.orbit4_blocked_zero	propext, Quot.sound	2026-09-26 transcript
Instances.Rule184.plateau_low_density	propext	2026-09-26 transcript
Instances.Rule184.plateau_has_jam	propext	2026-09-26 transcript
Instances.Rule184.plateau_jam_count_stalls	propext	2026-09-26 transcript
Instances.Rule184.low_density_four_one_step	propext, Classical.choice, Quot.sound	2026-09-26 transcript
Instances.Rule184Conservation.cars_conserved	propext, Quot.sound	2026-09-26 transcript
Instances.Rule184Small.clears_up_to_eight	propext, Classical.choice, Quot.sound	2026-09-26 transcript
Laws.G10LossfulPersistence.boundary_persistence	propext	2026-09-02 receipt
Laws.G10LossfulPersistence.checksum_lossful	none	2026-09-02 receipt
Laws.G10LossfulPersistence.checksum_protect	propext	2026-09-02 receipt
Laws.G10LossfulPersistence.checksum_regen	propext	2026-09-02 receipt
Laws.G10LossfulPersistence.checksum_boundary_persistence	propext	2026-09-02 receipt
Laws.G11GlobalAntiSymmetry.hierarchy_forces_aperiodic	none	2026-09-02 receipt
Laws.G11GlobalAntiSymmetry.g11_global_anti_symmetry	none	2026-09-02 receipt
Laws.G12FiniteWitnessRadiation.no_global_k_coloring	none	2026-09-02 receipt
Laws.G12FiniteWitnessRadiation.orbit_radiation	none	2026-09-02 receipt
Laws.G12FiniteWitnessRadiation.finite_witness_iff_no_global_coloring	none	2026-09-02 receipt
Laws.G13ThinOrbitSaturation.relative_density_theorem	propext	2026-09-02 receipt

Declaration	Axioms	Record
<code>Instances.ThinOrbit.power_members_iff</code>	propext, Quot.sound	2026-09-26 transcript
<code>Instances.ThinOrbit.power_count_sublinear</code>	propext, Classical.choice, Quot.sound	2026-09-26 transcript
<code>Instances.ThinOrbit.powers_infinite</code>	propext, Quot.sound	2026-09-26 transcript
<code>Instances.ThinOrbit.powers_relative_zero</code>	propext, Classical.choice, Quot.sound	2026-09-26 transcript
<code>Instances.ThinOrbit.counted_exceptions</code>	none	2026-09-26 transcript
<code>Instances.ThinOrbit.concrete_zero_density_failure</code>	propext, Classical.choice, Quot.sound	2026-09-26 transcript

S7.3 What each Lean statement covers

Table S8 records, for each law, which parts of the printed statement the Lean declarations establish and which are argued on paper or imported.

Table S8. Printed statements and their Lean coverage.

Law	In Lean	On paper or imported
G1	the reformulation; ghost discharge with both hypotheses; the Collatz ledger and split pairs, with the map defined on all natural numbers	the 2-adic completion and the ghost; oddness of the split pairs; the $2^L - 1$ family; the cycle condition
G2	no run is nonterminal at every step, in the native form comparing $T_k(x_k)$ with x_k	the printed conclusion that some x_k lies in A (classical extraction); Goodstein sequences; Kirby–Paris
G3	identity and bounds over integer costs; the binary counter	rational costs; the doubling array
G4	the three statements of Theorem 4.3; exhaustion and the finite leak for initial intervals	greedy unit fractions; Erdős–Straus results
G5	the no-go template; confinement; the four-phase family and the orbit of 22	the unbounded-horizon hypothesis for base 2
G6	covering from “a positive hole count later decreases”; hole formation from one-step propagation of non-visitation	the bridge from rate inequalities (open); windows and blocks; the designed variants
G7	the three parts, confinement in rank form; exclusion given a state in the escape invariant	other confinement arguments; the angel thresholds
G8	order independence of stable legal runs from a start with no infinite legal run; the two examples; the narrower least-action statement; the three-site sandpile	the least-action principle; sandpiles on arbitrary connected graphs
G9	the census bound for a sequence of natural numbers, in the form that the count is not unbounded on every tail; packaging of a supplied transporter record with one assigned defect; the rule-184 partial results of Proposition 6.11, including car conservation and evacuation for $n \leq 8$	the evacuation template beyond packaging; evacuation for every low-density ring (open); the defect ledger; the absorption hypothesis; the recurrence equality; rule 184
G10	the persistence theorem; the checksum cascade; Ducci nilpotency over $\mathbb{F}_2$; the integer Ducci collapse for $n = 4$	the integer collapse for other powers of two; Gilbreath

Law	In Lean	On paper or imported
G11	aperiodicity from the hierarchy and forced breaks, with admissibility and “breaks p ” as arbitrary predicates	finiteness of the regions R_p ; the tile sets
G12	restriction; orbit radiation; the equivalence for one k under the compactness hypothesis	the compactness theorem; the plane embedding of the Moser spindle, whose colouring obstruction is checked; the Hadwiger–Nelson witnesses
G13	the relative-density theorem, conditional on its two count hypotheses, with $o(\cdot)$ written as “for every j , eventually j times the count is at most” and c rational; the powers of two as an infinite missed family of density zero	the exponent bound; the application to a thick target; the reciprocity obstruction; all saturation theorems
Bridge	the underlying theorems for parts 1 and 2 and for the budgeted and eventually-zero cases of part 3	the status readings; the summable and contractive cases

S7.4 Hypotheses carried but not used

Table S9 lists the hypotheses that appear in a Lean statement and are not used by its proof.

Table S9. Unused hypotheses.

Declaration	Unused hypothesis
<code>Laws.G10LossfulPersistence.boundary_persistence</code>	the map is not injective
<code>Laws.G11GlobalAntiSymmetry.hierarchy_forces_aperiodic</code>	the zero period acts as the identity
<code>Laws.G3AmortizedCurrency.telescoping_bound</code> , <code>Laws.G3AmortizedCurrency.uniform_actual_cost_bound</code>	actual costs are nonnegative
<code>Laws.G80dometerAbelianization.least_action</code>	abelian compatibility; stability of the final configuration of the run whose counters are bounded

S8 Where the laws come from

Each law isolates the structure behind a problem that is easy to state and resists proof in a characteristic way: Collatz, Goodstein, Erdős–Straus, Lychrel, Recamán, the angel problem, sandpiles, Langton’s ant, Ducci, Gilbreath, aperiodic tilings, Hadwiger–Nelson and Apollonian packings. In each there is a tempting argument that fails; it relies on an auxiliary object — a measure on a completion, a fixed finite package, an ordinal — that is not part of the original system; and there is a certificate in the original system’s own terms that would make the argument valid. Each law differs from the laws of Tsiokos [116] by a quantifier over an unbounded run, scale or orbit.

Collatz and G1. The tempting argument is negative drift: on average the division exponent is 2, so $\log n$ changes by $\log 3 - 2 \log 2 < 0$ per step. This controls typical behaviour at best, and the conjecture concerns every orbit. The auxiliary object is the 2-adic completion, on which the exponent

statistics are exact and which contains non-descending points such as -1 . The certificate is the integer inequality $2^{A_k}n > 3^k n + B_k$ for some k , and G1 asks for such a certificate at every nonterminal point. Binary counters and size-change termination are solved relatives; aliquot sequences are an open one.

Sandpiles and G8. The tempting argument is that the order of topplings “obviously” does not matter; the auxiliary object is the final configuration, presumed unique before any hypothesis is named; the certificate is the vector of toppling counts. Abelian compatibility is the hypothesis that makes the argument valid (Theorem 4.11), and Example 4.13 shows that confluence alone does not give canonical counts. Rotor-routing is a solved relative.

Two related patterns that are not laws of this paper. Somos sequences divide at every step yet stay integral, because each term is a Laurent polynomial in a cluster-algebra structure; a non-descending auxiliary object whose selected consequence does descend is already a pattern of Tsiokos [116]. Conway’s look-and-say sequence eventually splits into a fixed finite set of elements; here the auxiliary object is a description of the system by itself, which falls within the self-reference studied in Tsiokos [113].

S9 Replay

All commands run from the repository root.

Lean library.

```
cd lean
lake build SixBirdsFoundationsVI
```

The axiom transcript of the instance theorems in the concordance is reproduced by

```
cd lean
lake env lean ../paper/supplement/instance_axioms.lean
```

Finite computations and their tests.

```
cd lab
pip install -e '[dev]'
pytest tests
python -m foundations_vi_lab.g01_solvency.run
python -m foundations_vi_lab.g01_solvency.run_l2
python -m foundations_vi_lab.g01_solvency.run_l3
python -m foundations_vi_lab.g01_solvency.run_l4
python -m foundations_vi_lab.g02_escrow.run
python -m foundations_vi_lab.g03_currency.run
python -m foundations_vi_lab.g04_exhaustion.run
python -m foundations_vi_lab.g05_confinement.run
python -m foundations_vi_lab.g06_needle.run
python -m foundations_vi_lab.g06_needle.run_l2
python -m foundations_vi_lab.g07_confinement.run
```

```
python -m foundations_vi_lab.g08_odometer.run all
python -m foundations_vi_lab.g09_transport.run
python -m foundations_vi_lab.g10_persistence.run
python -m foundations_vi_lab.g11_symmetry.run
python -m foundations_vi_lab.g12_finite_witness.run
python -m foundations_vi_lab.g13_saturation.run --bound 300000
```

Each runner prints its parameters with `-help`; with the options shown, its defaults are the parameters of the runs in Section S6, which are also stored with each run's output. The option `-write-results` saves the output. The package depends on `python-sat` and `sympy`.

Supplement tables and papers.

```
python3 paper/supplement/build_supp.py
cd paper
latexmk -pdf -interaction=nonstopmode -halt-on-error -outdir=build main.tex
latexmk -pdf -interaction=nonstopmode -halt-on-error -outdir=build supplement.tex
latexmk -pdf -interaction=nonstopmode -halt-on-error -outdir=build main.tex
```

The main paper and the supplement cite each other's numbered items, so the main paper is built a second time after the supplement.

References

- [1] Sheshansh Agrawal, Krishnendu Chatterjee, and Petr Novotný. Lexicographic ranking supermartingales: An efficient approach to termination of probabilistic programs. *Proceedings of the ACM on Programming Languages*, 2(POPL):34:1–34:32, 2018. doi:10.1145/3158122. arXiv:1709.04037.
- [2] Martin Aigner and Michael Fromme. A game of cops and robbers. *Discrete Applied Mathematics*, 8:1–12, 1984. doi:10.1016/0166-218X(84)90073-8.
- [3] Jean-Paul Allouche and Jeffrey Shallit. *Automatic Sequences: Theory, Applications, Generalizations*. Cambridge University Press, 2003.
- [4] Leonard Asimow and Ben Roth. The rigidity of graphs. *Transactions of the American Mathematical Society*, 245:279–289, 1978.
- [5] Michael Baake and Uwe Grimm. *Aperiodic Order. Volume 1: A Mathematical Invitation*. Cambridge University Press, 2013.
- [6] John M. Beggs and Dietmar Plenz. Neuronal avalanches in neocortical circuits. *Journal of Neuroscience*, 23(35):11167–11177, 2003.
- [7] Marc G. Bellemare, Sriram Srinivasan, Georg Ostrovski, Tom Schaul, David Saxton, and Rémi Munos. Unifying count-based exploration and intrinsic motivation. In *Advances in Neural Information Processing Systems 29 (NeurIPS 2016)*, 2016.
- [8] Robert Berger. *The Undecidability of the Domino Problem*, volume 66 of *Memoirs of the American Mathematical Society*. American Mathematical Society, 1966.
- [9] Elwyn R. Berlekamp, John H. Conway, and Richard K. Guy. *Winning Ways for your Mathematical Plays*, volume 2. Academic Press, 1982.
- [10] Allan Borodin and Ran El-Yaniv. *Online Computation and Competitive Analysis*. Cambridge University Press, 1998.
- [11] Allan Borodin, Jon Kleinberg, Prabhakar Raghavan, Madhu Sudan, and David P. Williamson. Adversarial queuing theory. *Journal of the ACM*, 48(1):13–38, 2001.
- [12] Jean Bourgain and Elena Fuchs. A proof of the positive density conjecture for integer Apollonian circle packings. *Journal of the American Mathematical Society*, 24(4):945–967, 2011. doi:10.1090/S0894-0347-2011-00707-8. arXiv:1001.3894.
- [13] Jean Bourgain and Alex Kontorovich. On the local-global conjecture for integral apollonian gaskets. *Inventiones Mathematicae*, 196:589–650, 2014.
- [14] Jean Bourgain and Alex Kontorovich. On zarembo’s conjecture. *Annals of Mathematics*, 180: 137–196, 2014.
- [15] Brian H. Bowditch. The angel game in the plane. *Combinatorics, Probability and Computing*, 16(3):345–362, 2007.
- [16] Joan Boyar, Lene M. Favrholdt, Michal Kotrbčík, and Kim S. Larsen. Relaxing the irrevocability requirement for online graph algorithms. *Algorithmica*, 84(7):1916–1951, 2022. Preliminary version in WADS 2017; arXiv:1704.08835.

- [17] Stephen Boyd, Arpita Ghosh, Balaji Prabhakar, and Devavrat Shah. Randomized gossip algorithms. *IEEE Transactions on Information Theory*, 52(6):2508–2530, 2006. doi:10.1109/TIT.2006.874516.
- [18] Florian Breuer. Ducci sequences in higher dimensions. *Integers*, 7:A24, 2007.
- [19] Florian Breuer, Ernest Lötter, and Brink van der Merwe. Ducci-sequences and cyclotomic polynomials. *Finite Fields and Their Applications*, 13(2):293–304, 2007.
- [20] Manuel Bronstein. *Symbolic Integration I*. Springer, 2005.
- [21] Kevin Brown. Digit reversal sums leading to palindromes, 2026. MathPages, <https://www.mathpages.com/home/kmath004/kmath004.htm>, accessed 2026-09-02.
- [22] Leonid A. Bunimovich and Serge E. Troubetzkoy. Recurrence properties of lorentz lattice gas cellular automata. *Journal of Statistical Physics*, 67(1-2):289–302, 1992.
- [23] John Canny. A computational approach to edge detection. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, PAMI-8(6):679–698, 1986. doi:10.1109/TPAMI.1986.4767851.
- [24] Marc Chamberland and David Thomas. The n-number ducci game. *Journal of Difference Equations and Applications*, 10(3):339–342, 2004. doi:10.1080/10236190410001647807.
- [25] Zachary Chase. A random analogue of Gilbreath’s conjecture. *Mathematische Annalen*, 388: 2611–2625, 2024. doi:10.1007/s00208-023-02579-w. arXiv:2005.00530.
- [26] Debashish Chowdhury, Ludger Santen, and Andreas Schadschneider. Statistical physics of vehicular traffic and some related systems. *Physics Reports*, 329:199–329, 2000. arXiv:cond-mat/0007053.
- [27] Edmund Clarke, Orna Grumberg, Somesh Jha, Yuan Lu, and Helmut Veith. Counterexample-guided abstraction refinement for symbolic model checking. *Journal of the ACM*, 50(5): 752–794, 2003. doi:10.1145/876638.876643.
- [28] John H. Conway. The angel problem. In *Games of No Chance*, volume 29 of *MSRI Publications*, pages 3–12. Cambridge University Press, 1996.
- [29] Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein. *Introduction to Algorithms*. MIT Press, 3rd edition, 2009.
- [30] Jean-François Coulombel and Grégory Faye. Sharp stability for finite difference approximations of hyperbolic equations with boundary conditions. *IMA Journal of Numerical Analysis*, 43(1): 187–224, 2023. doi:10.1093/imanum/drab088. arXiv:2102.03066.
- [31] George Cybenko. Dynamic load balancing for distributed memory multiprocessors. *Journal of Parallel and Distributed Computing*, 7(2):279–301, 1989.
- [32] Carlos F. Daganzo. The cell transmission model: A dynamic representation of highway traffic consistent with the hydrodynamic theory. *Transportation Research Part B*, 28:269–287, 1994.
- [33] Nicolaas G. de Bruijn and Paul Erdős. A colour problem for infinite graphs and a problem in the theory of relations. *Indagationes Mathematicae*, 13:369–373, 1951. Nederl. Akad. Wetensch. Proc. Ser. A 54.

- [34] Aubrey D. N. J. de Grey. The chromatic number of the plane is at least 5. *Geombinatorics*, 28(1):18–31, 2018. arXiv:1804.02385.
- [35] Rina Dechter. *Constraint Processing*. Morgan Kaufmann, 2003.
- [36] Nachum Dershowitz. Orderings for term-rewriting systems. *Theoretical Computer Science*, 17(3):279–301, 1982. doi:10.1016/0304-3975(82)90026-3.
- [37] Nachum Dershowitz and Zohar Manna. Proving termination with multiset orderings. *Communications of the ACM*, 22(8):465–476, 1979. doi:10.1145/359138.359142.
- [38] Deepak Dhar. Self-organized critical state of sandpile automaton models. *Physical Review Letters*, 64(14):1613–1616, 1990. doi:10.1103/PhysRevLett.64.1613.
- [39] Bruno Durand, Andrei Romashchenko, and Alexander Shen. Fixed point and aperiodic tilings. In *Developments in Language Theory (DLT 2008)*, volume 5257 of *Lecture Notes in Computer Science*, pages 276–288. Springer, 2008. arXiv:0802.2432.
- [40] Richard Durrett and David Griffeath. Asymptotic behavior of excitable cellular automata. *Experimental Mathematics*, 2(3):183–208, 1993. arXiv:patt-sol/9303002.
- [41] Larry Eisenberg and Thomas H. Noe. Systemic risk in financial systems. *Management Science*, 47(2):236–249, 2001.
- [42] Christian Elsholtz and Terence Tao. Counting the number of solutions to the erdos-raus equation on unit fractions. *Journal of the Australian Mathematical Society*, 94(1):50–105, 2013.
- [43] David Eppstein. Ten algorithms for egyptian fractions. *Mathematica in Education and Research*, 4(2):5–15, 1995.
- [44] Anne Fey, Lionel Levine, and Yuval Peres. Growth rates and explosions in sandpiles. *Journal of Statistical Physics*, 138:143–159, 2010. doi:10.1007/s10955-009-9899-6. arXiv:0901.3805.
- [45] Leonardo Fibonacci. Liber abaci. Manuscript; English translation: L. E. Sigler, *Fibonacci’s Liber Abaci*, Springer, 2002, 1202.
- [46] John G. Fletcher. An arithmetic checksum for serial transmissions. *IEEE Transactions on Communications*, COM-30(1):247–252, 1982.
- [47] Paul J. Flory. *Principles of Polymer Chemistry*. Cornell University Press, 1953.
- [48] Robert W. Floyd. Assigning meanings to programs. *Proceedings of Symposia in Applied Mathematics*, 19:19–32, 1967.
- [49] Thierry Fraichard and Hajime Asama. Inevitable collision states: A step towards safer robots? *Advanced Robotics*, 18(10):1001–1024, 2004. doi:10.1163/1568553042674662.
- [50] Henryk Fukś. Solution of the density classification problem with two cellular automata rules. *Physical Review E*, 55:R2081–R2084, 1997. arXiv:comp-gas/9703001.
- [51] Péter Gács. The angel wins, 2007. arXiv:0706.2817.
- [52] Martin Gardner. Mathematical games: The fantastic combinations of John Conway’s new solitaire game “life”. *Scientific American*, 223(4):120–123, October 1970.

- [53] Michael R. Garey, David S. Johnson, and Larry Stockmeyer. Some simplified np-complete graph problems. *Theoretical Computer Science*, 1:237–267, 1976. doi:10.1016/0304-3975(76)90059-1.
- [54] Gerhard Gentzen. Die widerspruchsfreiheit der reinen zahlentheorie. *Mathematische Annalen*, 112:493–565, 1936. doi:10.1007/BF01565428.
- [55] Allan Gibbard. Manipulation of voting schemes: a general result. *Econometrica*, 41:587–601, 1973.
- [56] R. L. Goodstein. On the restricted ordinal theorem. *Journal of Symbolic Logic*, 9(2):33–41, 1944.
- [57] James M. Greenberg and Stuart P. Hastings. Spatial patterns for discrete models of diffusion in excitable media. *SIAM Journal on Applied Mathematics*, 34(3):515–523, 1978.
- [58] Geoffrey Grimmett. *Percolation*. Springer, 2nd edition, 1999.
- [59] Bertil Gustafsson, Heinz-Otto Kreiss, and Joseph Oliger. *Time-Dependent Problems and Difference Methods*. Wiley, 2nd edition, 2013.
- [60] Richard K. Guy. *Unsolved Problems in Number Theory*. Springer, 3rd edition, 2004.
- [61] Richard K. Guy and John L. Selfridge. What drives an aliquot sequence? *Mathematics of Computation*, 29:101–107, 1975.
- [62] Summer Haag, Clyde Kertzer, James Rickards, and Katherine E. Stange. The local-global conjecture for apollonian circle packings is false. *Annals of Mathematics*, 200(2):749–770, 2024. arXiv:2307.02749.
- [63] Bernhard Haeupler and Amirbehshad Shahrashbi. Synchronization strings: Codes for insertions and deletions approaching the Singleton bound. In *Proceedings of the 49th Annual ACM SIGACT Symposium on Theory of Computing (STOC 2017)*, pages 33–46, 2017.
- [64] Marijn J. H. Heule. Computing small unit-distance graphs with chromatic number 5. *Combinatorics*, 28(1):32–50, 2018. arXiv:1805.12181.
- [65] Nicholas J. Higham. *Accuracy and Stability of Numerical Algorithms*. SIAM, 2nd edition, 2002.
- [66] Alexander E. Holroyd, Lionel Levine, Karola Mészáros, Yuval Peres, James Propp, and David B. Wilson. Chip-firing and rotor-routing on directed graphs. In *In and Out of Equilibrium 2*, volume 60 of *Progress in Probability*, pages 331–364. Birkhäuser, 2008. arXiv:0801.3306.
- [67] Shlomo Hoory, Nathan Linial, and Avi Wigderson. Expander graphs and their applications. *Bulletin of the American Mathematical Society*, 43:439–561, 2006.
- [68] Robert Hough. Solution of the minimum modulus problem for covering systems. *Annals of Mathematics*, 181(1):361–382, 2015.
- [69] Emmanuel Jeandel and Michaël Rao. An aperiodic set of 11 Wang tiles. *Advances in Combinatorics*, 2021(1):1–37, 2021. doi:10.19086/aic.18614. arXiv:1506.06492.
- [70] Harry Kesten. *Percolation Theory for Mathematicians*. Birkhäuser, 1982.

- [71] Laurie Kirby and Jeff Paris. Accessible independence results for peano arithmetic. *Bulletin of the London Mathematical Society*, 14(4):285–293, 1982.
- [72] Oddvar Kloster. A solution to the angel problem. *Theoretical Computer Science*, 389(1-2): 152–161, 2007.
- [73] Paul C. Kocher. Timing attacks on implementations of Diffie-Hellman, RSA, DSS, and other systems. In *Advances in Cryptology — CRYPTO '96*, volume 1109 of *Lecture Notes in Computer Science*, pages 104–113. Springer, 1996.
- [74] Alex Kontorovich and Hee Oh. Almost prime pythagorean triples in thin orbits. *Journal für die reine und angewandte Mathematik*, 667:89–131, 2012.
- [75] Jeffrey C. Lagarias. The $3x+1$ problem and its generalizations. *American Mathematical Monthly*, 92(1):3–23, 1985. doi:10.2307/2322189.
- [76] Jeffrey C. Lagarias, editor. *The Ultimate Challenge: The $3x+1$ Problem*. American Mathematical Society, 2010.
- [77] Gerard Laman. On graphs and rigidity of plane skeletal structures. *Journal of Engineering Mathematics*, 4:331–340, 1970. doi:10.1007/BF01534980.
- [78] Christopher G. Langton. Studying artificial life with cellular automata. *Physica D*, 22:120–149, 1986.
- [79] Steven M. LaValle. *Planning Algorithms*. Cambridge University Press, 2006.
- [80] Chin Soon Lee, Neil D. Jones, and Amir M. Ben-Amram. The size-change principle for program termination. *ACM SIGPLAN Notices*, 36(3):81–92, 2001. POPL 2001.
- [81] Daniel Liberzon. *Switching in Systems and Control*. Birkhäuser, 2003.
- [82] M. J. Lighthill and G. B. Whitham. On kinematic waves. ii. a theory of traffic flow on long crowded roads. *Proceedings of the Royal Society A*, 229:317–345, 1955.
- [83] Douglas Lind and Brian Marcus. *An Introduction to Symbolic Dynamics and Coding*. Cambridge University Press, 1995.
- [84] Neal Madras and Gordon Slade. *The Self-Avoiding Walk*. Birkhäuser, 1993.
- [85] András Máthé. The angel of power 2 wins. *Combinatorics, Probability and Computing*, 16(3): 363–374, 2007.
- [86] Louis J. Mordell. *Diophantine Equations*, volume 30 of *Pure and Applied Mathematics*. Academic Press, 1969.
- [87] Kai Nagel. Particle hopping models and traffic flow theory. *Physical Review E*, 53:4655–4672, 1996. arXiv:cond-mat/9509075.
- [88] Richard Nowakowski and Peter Winkler. Vertex-to-vertex pursuit in a graph. *Discrete Mathematics*, 43:235–239, 1983. doi:10.1016/0012-365X(83)90160-7.
- [89] Andrew M. Odlyzko. Iterated absolute values of differences of consecutive primes. *Mathematics of Computation*, 61(203):373–380, 1993.

- [90] OEIS Foundation Inc. A005132, 2026. The On-Line Encyclopedia of Integer Sequences, <https://oeis.org/A005132>, accessed 2026-09-02.
- [91] OEIS Foundation Inc. A057167, 2026. The On-Line Encyclopedia of Integer Sequences, <https://oeis.org/A057167>, accessed 2026-09-02.
- [92] OEIS Foundation Inc. A060382, 2026. The On-Line Encyclopedia of Integer Sequences, <https://oeis.org/A060382>, accessed 2026-09-02.
- [93] Omur Ozel, Kaya Tutuncuoglu, Jing Yang, Sennur Ulukus, and Aylin Yener. Transmission with energy harvesting nodes in fading wireless channels: Optimal policies. *IEEE Journal on Selected Areas in Communications*, 29(8):1732–1743, 2011. arXiv:1106.1595.
- [94] Jaan Parts. Graph minimization, focusing on the example of 5-chromatic unit-distance graphs in the plane. *Geombinatorics*, 29(4):137–166, 2020. arXiv:2010.12665.
- [95] Roger Penrose. The role of aesthetics in pure and applied mathematical research. *Bulletin of the Institute of Mathematics and its Applications*, 10:266–271, 1974.
- [96] Marcus Pivato. Defect particle kinematics in one-dimensional cellular automata. *Theoretical Computer Science*, 377(1–3):205–228, 2007. doi:10.1016/j.tcs.2007.03.014. arXiv:math/0506417.
- [97] Simon Plouffe. Verification of gilbraith’s conjecture up to 10^{14} , 2025. arXiv:2510.06688.
- [98] Paul I. Richards. Shock waves on the highway. *Operations Research*, 4:42–51, 1956. doi:10.1287/opre.4.1.42.
- [99] Robert H. Risch. The problem of integration in finite terms. *Transactions of the American Mathematical Society*, 139:167–189, 1969.
- [100] Neil Robertson and Paul D. Seymour. Graph minors. xx. wagner’s conjecture. *Journal of Combinatorial Theory, Series B*, 92:325–357, 2004. doi:10.1016/j.jctb.2004.08.001.
- [101] Raphael M. Robinson. Undecidability and nonperiodicity for tilings of the plane. *Inventiones Mathematicae*, 12:177–209, 1971. doi:10.1007/BF01418780.
- [102] Mark A. Satterthwaite. Strategy-proofness and Arrow’s conditions: existence and correspondence theorems for voting procedures and social welfare functions. *Journal of Economic Theory*, 10:187–217, 1975.
- [103] Marjorie Senechal. *Quasicrystals and Geometry*. Cambridge University Press, 1995.
- [104] Dan Shechtman, Ilan Blech, Denis Gratias, and John W. Cahn. Metallic phase with long-range orientational order and no translational symmetry. *Physical Review Letters*, 53(20):1951–1954, 1984. doi:10.1103/PhysRevLett.53.1951.
- [105] Daniel D. Sleator and Robert E. Tarjan. Self-adjusting binary search trees. *Journal of the ACM*, 32(3):652–686, 1985.
- [106] David Smith, Joseph Samuel Myers, Craig S. Kaplan, and Chaim Goodman-Strauss. An aperiodic monotile. *Combinatorial Theory*, 4(1):#6, 2024. doi:10.5070/C64163843. arXiv:2303.10798.

- [107] David Smith, Joseph Samuel Myers, Craig S. Kaplan, and Chaim Goodman-Strauss. A chiral aperiodic monotile. *Combinatorial Theory*, 4(2):#13, 2024. doi:10.5070/C64264241. arXiv:2305.17743.
- [108] Jonathan Stone, Michael Greenwald, Craig Partridge, and James Hughes. Performance of checksums and crcs over real data. *IEEE/ACM Transactions on Networking*, 6(5):529–543, 1998.
- [109] John C. Strikwerda. *Finite Difference Schemes and Partial Differential Equations*. SIAM, 2nd edition, 2004.
- [110] James Joseph Sylvester. On a point in the theory of vulgar fractions. *American Journal of Mathematics*, 3(4):332–335, 1880.
- [111] Terence Tao. Almost all orbits of the Collatz map attain almost bounded values. *Forum of Mathematics, Pi*, 10:e12, 2022. doi:10.1017/fmp.2022.8. arXiv:1909.03562.
- [112] Robert E. Tarjan. Amortized computational complexity. *SIAM Journal on Algebraic and Discrete Methods*, 6(2):306–318, 1985. doi:10.1137/0606031.
- [113] Ioannis Tsiokos. Six birds foundations V: Endogenous closure. A catalog of structural laws for living, cognitive, and social systems, 2026.
- [114] Ioannis Tsiokos. Audited operational realisability: A closure-completion reflection for carriers with audit data, 2026.
- [115] Ioannis Tsiokos. Six birds: Foundations of emergence calculus, 2026.
- [116] Ioannis Tsiokos. Six birds foundations iv: A catalog of layer-agnostic structural laws, 2026.
- [117] Ioannis Tsiokos. To flatten a stone with six birds: Critical pairs, holonomy, and confluence in rewriting systems, 2026.
- [118] R. C. Vaughan. On a problem of Erdős, Straus and Schinzel. *Mathematika*, 17(2):193–198, 1970.
- [119] John Walker. Three years of computing: Final report on the palindrome quest, 1990. Fourmilab, <https://www.fourmilab.ch/documents/threeyears/>, accessed 2026-09-02.
- [120] Jan C. Willems. Dissipative dynamical systems, part i: General theory. *Archive for Rational Mechanics and Analysis*, 45:321–351, 1972. doi:10.1007/BF00276493.
- [121] Andrew P. Witkin. Scale-space filtering. In *Proceedings of the 8th International Joint Conference on Artificial Intelligence (IJCAI-83)*, pages 1019–1022, 1983.
- [122] Jing Yang and Sennur Ulukus. Optimal packet scheduling in an energy harvesting communication system. *IEEE Transactions on Communications*, 60(1):220–230, 2012. arXiv:1010.1295.